1 Introduction

What makes a conversation system dynamic as opposed to static? In the early days of formal conversational analysis, Stalnaker (1978), Karttunen (1976), Lewis (1979), Kamp (1981), and Heim (1983) gave formal characterizations of how interlocutors track conversational contributions, and provided constraints on how incoming utterances might rely on previous ones. These analyses proliferated in response to aspects of meaning that (seemingly) extended beyond the bounds of a single sentence. Some took the semantic contribution of an utterance, not to be a static proposition (in the tradition of Montague (1970)), but rather the utterance's *context change potential* (CCP). On these accounts, the meaning of an utterance was inextricably tied to how the utterance would impact the conversation. Frameworks that took an utterance's semantic contribution to be a CCP became classically known as "dynamic" semantic theories. In contrast, other theories posited only a "static" semantics, leaving any conversational effects of an utterance to pragmatics.

Is there any deep difference between static and dynamic theories of conversation? The classic answer, given by van Benthem (1986, 1996) (hereafter, vB), is that a theory of conversation is static when the system it specifies is ELIMINATIVE and DISTRIBUTIVE. However, Rothschild & Yalcin (2015, 2016) (hereafter, R&Y) propose an alternative: (strongly) static conversation systems are IDEMPOTENT and COMMUTATIVE. R& Y present strong staticness as a conservative extension of vB: it applies to a wider class of systems (i.e. incremental ones), while agreeing with vB wherever his theorem applies. This is incorrect. R&Y's criterion diverges systematically from vB's within his own domain - systems whose conversational states are arbitrary sets of possible worlds.

I argue this in three steps. Simple Veltman (Section 4) is an exemplar of the divergence: it is static by R&Y's lights, but dynamic by van Benthem's. Section 5 shows that this divergence holds across an entire class. Every *test collapse* operator (definition XX) is idempotent and commutative regardless of what is tested, and vB dynamic so long as the test is non-trivial. Section 6 explains why this happens: vB's criterion asks whether an operation is intersective using the system's own worlds, while R&Y's definition demonstrably more permissive as it merely asks whether some intersective system (which can be built out of higher-type objects, with states standing in for worlds) is isomorphic to the system in question. These results are two pieces are of one argument about R&Y's system: the generality that their characterization has is purchased by the permissiveness their theoretical mechanism.

2 Characterizing Staticness

Before specifying vB and R&Y's formal characterizations of the static/dynamic divide, I first give intuitive characterizations of prototypical static and dynamic conversation systems using a simplified version of Stalnaker (1978). From these intuitive notions, we can extract the desiderata identified by R&Y (2015).

2.1 Static Models of Conversation

Stalnaker (1978) and Lewis (1979) both give classic frameworks for modelling discourse. They share two observations about conversation: (i) what someone asserts often departs from the literal meaning of what they say, and that (ii) conversation evolves in a rule-governed way. From these observations, Lewis and Stalnaker claim that the rules constraining conversation come from more general constraints on rationality, permissibility, etc, and that we can and should accommodate the gap between literal and asserted content by appeal to these general constraints.

On Stalnaker's system particularly, an assertion expresses a *proposition* - a set of possible worlds - against a *conversational context* which is also a set of possible worlds. The static meaning of an utterance is the proposition with which it is associated. The act of making an assertion prompts updating the proposition associated with that utterance to the context. This update consists in intersecting the proposition expressed with the current context to deliver a new conversational context.

(1) Stalnaker Assertion Rule : $c[\phi^k] := c \cap [\![\phi]\!]^k$

The assertion rule models the characteristic pragmatic **conversational** effect of *assertion*. Stalnaker (1978) is then the prototypical example of a static conversation system because it pairs a static mapping from utterances to truth-conditions with a dynamic pragmatic assertion rule. The mapping from utterance to truth-condition is static because it does not update the context; the pragmatic assertion rule is dymanic because it does.

2.2 Dynamic Models of Conversation

What about a prototypical dynamic system? Though there is (perhaps) less consistency across dynamic systems, they generally have a similar broad structure. They take the semantic contribution of a sentence to be the effect an utterance of it has on the context, i.e., a *context change potential*. In other words, unlike in a Stalnakerian system where all of the effort lies in characterizing the proposition to be updated, dynamic systems take the work to be characterizing the update itself.

So, how should we intuitively draw the divide between static and dynamic systems? One orthodox way (von Fintel and Gillies (2007) from van Benthem (1986, 1989)), is that a system is not interestingly dynamic if it is "translatable" into a static system. We will say that a dynamic system is *translatable* into a static one iff for every CCP in the language, there is a possible proposition, which when intersected with the context has the same effect as updating with that CCP. Formally, we can model this using the simple intersective rule.

(2) Simple Intersective Rule : $c[\phi] := c \cap [\![\phi]\!]$

Notice the similarity to the stalnakerian assertion rule. A system, then, is *trivially dynamic* just in case there is a set of static propositions that when paired with the simple intersective rule generate precisely the same updates as the dynamic system's set of CCPs.

3 Formalizing Conversational Models

In order to formally specify what does and does not count as a translation between conversational systems, we need a formal definition of what counts as a *conversation system* - the sort of thing that can or cannot be static/dynamic. Rothschild and Yalcin (2016) do this by distinguishing between a *conversation system* and a *state system*.

(3) Conversation System : a triple $\langle L, C, \cdot[\cdot]\rangle$, where L is a set of sentences, C is a set of conversational states, and $\cdot[\cdot]$ is an *update function* from L to a set of context change potentials (unary operations) on C (in other words $\cdot[\cdot] : L \rightarrow (C \rightarrow C)$).

(4) State System : a pair of a set C of conversational states and a set O of unary operations (context change potentials) o on C (so that $o : C \rightarrow C$ for every $o \in O$).

Crucially, R&Y (2015) point out that the intuitive idea we gave for a static system can be formally modeled as any conversational system which has an intersective state system.

(5) Conversation System Staticness : A conversation system such that (i) every sentence ϕ is associated with a set of points $[\![\phi]\!]$, (ii) every conversational state is given by a set of points, and (iii) update goes via the simple intersective rule.

(6) A state system $\langle C, O\rangle$ is *intersective* just in case $C \subseteq \wp(W)$ for some set W, and there exists some set $P \subseteq \wp(W)$ such that for all $o \in O$, there exists $p \in P$ such that for any $c \in C$, $co = c \cap p$. (R&Y (2015))

R&Y introduce an additional component into this mix. They subsequently define strong staticness more generally for any conversation system, as one which is isomorphic to an intersective conversation system.

(7) Strong Staticness: A conversation system $\langle L, C, \cdot[\cdot]\rangle$ is strongly static just in case there exists an intersective conversation system $\langle L, C\prime, \cdot[\cdot]\prime\rangle$ and a bijection $f : C \rightarrow C\prime$ such that $f(c[\phi]) = f(c)[\phi]\prime$ for every $\phi \in L$.

Contrast this with the initial definition of an intersective state system, definition (6): intersectivity is a property of the given $\langle C, O\rangle$, checked using the system's own set of worlds, W, with no bijection anywhere in the definition. Strong staticness is existential — $C\prime$ need not be built over the pre-existing W.

We have two ways of getting to staticness on board - either conversation systems whose states and operations can be defined merely via points and the intersective rule, or as conversation systems which are isomorphic to an intersective conversation system. With this formal background ,we can now ask - is there a set of properties which pick out all and only the intersective state systems? This is precisely the question to which van Benthem's and Rothschild & Yalcin's characterizations are answers.

3.1 Van Benthem

Van Benthem showed that intersective state systems (and consequently the "static" conversation systems) are characterized by two properties: ELIMINATIVITY and DISTRIBUTIVITY.[1] Formally:

(8) INTERSECTIVITY: A state system $\langle \wp(W), O\rangle$ is intersective iff for all $c, c' \in \wp(W)$ and $o \in O$,

a. ELIMINATIVITY: $co \cup c = c$

b. FINITE DISTRIBUTIVITY: $(c \cup c')o = co \cup c'o$

In other words, for a semantics to be truly dynamic, it has to be the case that either update happens such that the resulting set of conversation states is not a subset of the input, or updating some proposition point-wise to the input set is not equivalent to updating the proposition to the set of conversational states all at once.

Consider a simple version of Veltman's (1996) semantics for epistemic modals, , from von Fintel and Gillies (2007), (I will refer to this as EMBEDDABLE VELTMAN). The syntax is identical to the syntax for propositional logic, plus the $\diamond$ operator (i.e., if ϕ is a WFF, then $\diamond\phi$ is a WFF). The semantics is below.

(9) **"EMBEDDABLE" VELTMAN**

$$s[p] \coloneqq s \cap [\![p]\!]$$

$$s[\phi \wedge \psi] \coloneqq s[\phi][\psi]$$

$$s[\neg\phi] \coloneqq s - s[\phi]$$

$$s[\diamond\phi] \coloneqq \begin{cases} s[\phi] \neq \emptyset & \text{return s} \\ s[\phi] = \emptyset & \text{return } \emptyset \end{cases}$$

Notice that the clause for $\diamond\phi$ here makes for a non-distributive (but eliminative) language. Suppose $s =$

[1] This result requires that the space of conversation states for the state system be a powerset of a given set of worlds, W. i.e. $C = \wp(W)$ for some W.

$\{w_1, w_2\}$ and $[\![\phi]\!] = \{w_1\}$. Here, distributivity fails because updating s wholesale results in a different set of final worlds than updating the individual worlds in s point-wise with $[\![\diamond\phi]\!]$, and adding the results.

(10) WHOLESALE UPDATE: $s[\diamond\phi] = s$

(11) POINTWISE UPDATE: $w_1[\diamond\phi] \cup w_2[\diamond\phi] = \{w_1\} \cup \emptyset$
$= \{w_1\}$

Pointwise update leads to a different result than updating the whole context at once – hence distributivity fails. However, this semantics is still eliminative because s can only ever be reduced; it can't "grow" in size.[2,3]

3.1.1 Limitations of van Benthem

Recall that we defined a static conversation system as a conversation system which is associated with an intersective state system. This assumption is crucial for vB's characterization of staticness. However, R&Y (2015) suggest that the notion of staticness should be applicable beyond just intersective state systems because the classic Stalnakerian "static" semantics can be reframed in non-intersective terms. For example, one might propose that static update happens INCREMENTALLY – updating a proposition to the context is simply a matter of set-theoretic union between a context (in this case, a set of propositions) and the singleton set containing the new proposition.

(12) INCREMENTALITY : A state system $\langle C, O\rangle$ is *incremental* just in case for some set P, $C \subseteq \wp(P)$, and for all $o \in O$, there exists $p \in P$ such that $co = c \cup \{p\}$ for any $c \in C$.

R&Y emphasize that these incremental state systems seem intuitively static. But given vB, we wouldn't be able to say whether they're static or dynamic because they are non-intersective.

3.2 Rothschild and Yalcin

With this worry in mind, R&Y provide a definition of staticness that applies to both intersective and incremental systems. They show that a state system is *strongly static* (i.e., isomorphic to an intersective state system) iff it is both IDEMPOTENT and COMMUTATIVE.

(13) STRONG STATICNESS : A state system $\langle C, O\rangle$ is strongly static iff for all $o, o' \in O$ and $c \in C$,

[2]For a more complete explanation see von Fintel and Gillies (2007).

[3]Groenendijk and Stokhof's (1991) DPL is a classic example of a non-eliminative, but distributive system.

a. IDEMPOTENCE : $coo = co$

b. COMMUTATIVITY : $coo' = co'o$

By their definition, DPL and "usual versions of update semantics (Veltman 1996)", particularly EMBEDDABLE VELTMAN are not strongly static, since their systems are neither commutative nor idempotent. Recall EMBEDDABLE VELTMAN from section IIIa, and let's apply R&Y's definitions to it.

(14) **EMBEDDABLE VELTMAN**

$$
\begin{aligned}
s[p] &:= s \cap [\![p]\!] \\
s[\phi \wedge \psi] &:= s[\phi][\psi] \\
s[\neg\phi] &:= s - s[\phi] \\
s[\diamond\phi] &:= \begin{cases} s[\phi] \neq \emptyset & \text{return s} \\ s[\phi] = \emptyset & \text{return } \emptyset \end{cases}
\end{aligned}
$$

This semantics is neither idempotent nor commutative. Epistemic contradiction $\diamond\phi \wedge \neg\phi$ is one case where the semantics fails idempotency. Updating with the epistemic contradiction once tests to make sure the original state had some ϕ-worlds, and then throws all of those out. Updating the output state with the epistemic contradiction again necessarily results in the absurd state. Why isn't *Embeddable Veltman* commutative? Because changing the order of update changes the output. So, R&Y characterize *Embeddable Veltman* in line with vB's characterization.

They conclude then that vB's characterization using eliminativity and distributivity is "sufficient but not necessary" for strong staticness:

> One simple way to see this is to note that van Benthem's result requires the set of conversational states to have Boolean structure, whereas there are strongly static conversation systems which do not have Boolean structure..."

So - even system from three states $\{\{1\},\{0\},\{\emptyset\}\}$ with two operations, a and b which represent literal intersections with {1}, and {0}, indicated in Figure 1 below, can not be evaluated using van Benthem's characterization, because the system isn't Boolean. So their primary stated motivation for introducing their analysis is domain-extension cases — van Benthem's theorem does not apply in the case illustrated by Figure 1 because C is not a full powerset.

4 Considerations

So far everything looks great for R&Y, they carve the existing landscape the way vB does, and their analysis extends to capture non-intersective conversation systems. Nonetheless, there is one upshot to note here about the two ways of carving dynamicness.

First, a slight modification to our semantics for modals that prohibits embeddings of $\diamond$ will render it idempotent[4]; I'll call this system *Original Veltman*.[5] While this doesn't result in characterizing Original Veltman as static, it is one initial step towards a "static" version of Veltman (1996), which I'll introduce later as *Simple Veltman*.

Veltman (1996) originally set up the syntax of his language such that $\diamond$ could not be embedded under the Boolean operators. This non-embeddability comes from associating two languages, $L_0^{\mathbf{A}}$ $L_1^{\mathbf{A}}$ with a set of finitely many atomic sentences, **A**.

vB's results reflect an intuition that these questions are separate - both *Embeddable Veltman* and *Original Veltman* are vB-dynamic because they are non-distributive. Alternatively, though *Embeddable Veltman* is NON-IDEMPOTENT, *Original Veltman* **is** trivially IDEMPOTENT due to the non-embeddability of *might*.

4.1 *Simple Veltman*

Consider a system analgous to *Original Veltman*, but highly restricted. Its only operation tests whether the state is compatible with a given proposition, p, and then returns either the empty state or input state (in accordance with Veltman's *might*-operator). Such a system is a static semantics according to R&Y's characterization, but vB dynamic. Suppose the following syntax and semantics.

(15) **SIMPLE VELTMAN (SV)**

$L_{\mathbf{SV}}$ Syntax

If ϕ is an atomic formula of $L_{\mathbf{SV}}$, then ϕ is a WFF.

If ϕ is an atomic formula of $L_{\mathbf{SV}}$, then $\ulcorner\diamond\phi\urcorner$ is a WFF.

$L_{\mathbf{SV}}$ Semantics

$$s[\diamond\phi] := \begin{cases} s \cap [\![\phi]\!] \neq \emptyset & \text{return s} \\ s \cap [\![\phi]\!] = \emptyset & \text{return } \emptyset \end{cases}$$

[4]Though it remains the same under vB's characterization - non-distributive and eliminative.

[5]This modification is actually how Veltman (1996) is stated, although recapitulations of Veltman (1996) (i.e., EMBEDDABLE VELTMAN) often allow $\diamond$ to scope anywhere, see von Fintel and Gillies (2007).

A note about SV - atomic formulas are well-formed in the language of Simple Veltman, but the semantics assigns them no update clause of their own — only $\diamond$-formulas are updatable. This is not an oversight; Section 5 demonstrates it is exactly what keeps Simple Veltman's operation of the proper form for the divergence, such that idempotence and commutativity hold independently of content. If you assign atomic formulas an ordinary intersective clause instead, this produces FACTS+CHECKS (given below) , which is not strongly static.

$\mathrm{L}_{\mathbf{SV}}$ is both idempotent and commutative, because there is no sentence or sequence of updates definable in this language for which commutativity or idempotence fails. We will see this illustrated in more detail below. The sentences we relied on for failures of commutativity and idempotence in *Embeddable Veltman* (i.e., epistemic contradictions) required being able to define '$\neg$' and/or '$\wedge$' in our semantics. Since this is no longer possible because these operators don't exist in the $\mathrm{L}_{\mathbf{SV}}$, the semantics is commutative and idempotent.

Let's walk through why we cannot define a non-commutative or non-idempotent update for $\mathrm{L}_{\mathbf{SV}}$ given two independent propositions p and q. There are four worlds in W to consider:

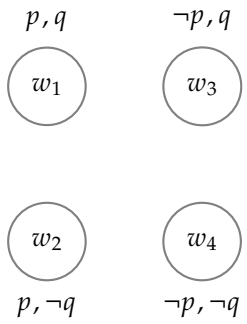


Figure 1: Possibilities for p and q

$\mathrm{L}_{\mathbf{SV}}$ is commutative because there is no set within $\wp(W)$, to which we could apply $\diamond p$ or $\diamond q$ where order changes the output. Consider a simple case like $\{w_1, w_2\}[\diamond p][\diamond q]$ as opposed to $\{w_1, w_2\}[\diamond q][\diamond p]$. In the first case, $[\![p]\!] = \{w_1, w_2\}$ and so its intersection with $\{w_1, w_2\}$ just returns the initial state. However, $[\![q]\!] = \{w_1, w_3\}$, and so its intersection with the initial state returns $\{w_1\}$ which is a non-empty state. Considering $\ulcorner\diamond\phi\urcorner$, any value of ϕ, which is compatible with the initial state, will return the initial state no matter what order it is updated in, and any q, which is not compatible with the initial state, will return the empty set, which will then continually be returned for any other ϕ no matter what $\diamond\phi$ you updated with.[6]

What about IDEMPOTENCE? In order for the system to be NON-IDEMPOTENT, we need to define an expression such that updating with it twice differs from updating with it once. W[$\diamond p$] = W, and W[$\diamond p$][$\diamond p$]=W. Further

[6]Because there is no ϕ, which when updated will return a non-empty state given an empty initial state.

any subset of $\wp(W)$ that $\diamond p$ gets updated with will either return that subset or the empty set. In either case, once the s is returned, continual updating with the same formula will return that same state, similarly if a first update with $\diamond\phi$ returns the empty state, subsequent updates will never return a non-empty state. Hence, $\mathrm{L}_{\mathbf{SV}}$ is idempotent.

Since $\mathrm{L}_{\mathbf{SV}}$ is IDEMPOTENT and COMMUTATIVE, it is *static* for R&Y. It contains no update that is order-sensitive, and no expression impacts the context differently depending on how many times it is updated. The only "expressive" power the system has is to check and see what the current context is compatible with. Hence, from a top-down view of the system, the $\diamond$-operator looks inert.

In contrast, $\mathrm{L}_{\mathbf{SV}}$ is dynamic for vB because it fails to be distributive for the same reason that *Embeddable Veltman* and *Original Veltman* are — because the semantics for $\diamond\phi$ treats updates globally (given an input, it returns the whole input or $\emptyset$). So, the results of pointwise update with $\diamond\phi$ can come apart from global update of $\diamond\phi$.

Note further that a non-distributive system is the only way for a state system to be vB dynamic and Y&S static. This is because non-eliminativity is incompatible with idempotence.

Assume for sake of contradiction there is an operator $\cdot[\cdot]$ which has the following two properties:

$$\begin{aligned} &\text{For some } \phi,\ c[\phi] \not\subseteq c && \textsc{non-eliminativity} \\ &\text{For all } \phi,\ c[\phi][\phi] = c[\phi] && \textsc{idempotence} \end{aligned} \tag{16}$$

By non-eliminativity, there will be some ϕ such that $c[\phi][\phi] \not\subseteq c[\phi]$, and definitionally $c[\phi][\phi] \neq c[\phi]$. But contradiction, since by assuming idempotence we assumed that for every sentence ϕ, $c[\phi][\phi] = c[\phi]$. Hence, NON-ELIMINATIVITY $\Rightarrow$ NON-IDEMPOTENCE.

This simple-minded illustration shows that there is no way to develop a system which is idempotent, and non-eliminative, which would in principle be Y&S static, but vB dynamic. This is worthy of note because it shows that the pull-apart that this example takes advantage of is largely reliant on the way that distributivity can fail independent of the other three properties.[7]

[7]Further exploration can be done to show that any update which renders a system non-commutative is compatible with distributivity and eliminativity. (e.g. that if a system is intersective and vB static, it will not be classified as Y&R dynamic).

5 Generalizing the Witness

A natural knee-jerk reaction to Simple Veltman is that it only receives an R&Y verdict of strongly static because it is deficient. Atomic formulas have no update clause in the semantics, so there is almost nothing there to fail idempotence or commutativity. However, what produces the verdict is the shape of the operation, not the poverty of the language. This finding generalizes to any test of this form whatsoever.

First, a few ground-clearing definitions. I define the set of operations on which R&Y and vB diverge as test-collapse operators which take an input context to a fixed point.

(17) *Fixed Point*: A state c is a fixed point of an operation $o \in O$ just in case $co = c$

(18) Test-collapse Operator: For a predicate B on $\wp(W)$ and a fixed $K \subseteq W$, the operation $o_{B,K}$ is defined as: $co_{B,K} = c$ if $B(c)$, and $co_{B,K} = K$ otherwise

Note that Veltman's $\diamond\phi$ is a test collapse operator, which returns c, when ϕ is true and otherwise returns a degenerate value K, here the empty state. In other words:

(19) B(c) := $c \cap [\![\phi]\!] \neq \emptyset, K = \emptyset$

Note that it is fully licensed under this definition of a compatibility test for the proposition, p, that is intersected with the context,c ϕ = W or $\phi = \emptyset$. At ϕ = W, the test reduces down to an identity mapping, it returns the original context. At $\phi = \emptyset$, the compatibility test is uniformly false, and the operation is a constant map to $\emptyset$.[8]

First, for every predicate B on $\wp(W)$ and every $K \subseteq W$, the operation $o_{B,K}$ is idempotent (proof in the appendix). The idea here is that the operation has only two possible outputs: the input state c, or the collapse state, K. Both are states that subsequent applications of o leaves alone - c because passing the test means being handed back unchanged, and K because it returns itself as a fixed point. The result is perfectly straightfoward. SV's idempotence is not a fact about epistemic might or Veltman's syntax, it is a fact about the shape of consistency operators.[9]

[8]Both of these instances are genuinely intersective within their conversation systems (a la vB) and so are distributive, and hence static systems on vB's definition.

[9]This is dependent on multiple tests in a system having the same collapse state!!!

5.1 Variation of the Predicate and the Collapse State

However, vB's analysis is sensitive to the collapse state. Van Benthem's criterion constrains the collapse target sharply. Specifically:

(20) For every predicate B on $\wp(W)$, let $F_B := \{c \in \wp(W) : \neg B(c)\}$ be the states on which B fails. Then $o_{B,K}$ is eliminative if and only if $K \subseteq \bigcap F_B$[10]

The idea here is that eliminativity demands that the output always sit inside the input. So, eliminativity will fail, not where states pass the test, but where they fail because the only times the operation changes the state from the input state is when the test is failed, and the output is K. So, eliminativity reduces to a single demand in this instance: K must fit inside every state that fails the test; in other words, be a subset of their intersection. The formal proof in the appendix is just this observation run in both directions.

This means that van Benthem's constraint of eliminativity often has a strong opnion about where a failed test may send you. This is especially apparent for compatibility tests. Under vB's analysis, compatibilty tests require $K = \emptyset$.

(21) If $B(\emptyset)$ is false[11], then $o_{B,K}$ is eliminative if and only if K = $\emptyset$.

This again is straightforward. Supposing that the empty state is one on which B fails, then the empty set is a member of F_B by definition. But then the intersection of sets in F_B must itself be $\emptyset$ by definition. However, we just showed that $o_{B,K}$ is eliminative iff $K \subseteq F_B$. So, $K = \emptyset$.

The key here is that the absurd state as a consequence of failing a might-test is not a design choice under vB's analysis, but a consequence. Compatibility tests, as described in the literature, that are eliminative must result must use the absurd state as their fixed point.

More importantly, however, is the fact that non-distributivity is a general feature of non-vacuous test-collapse operators.

(22) Fix $K = \emptyset$. For B$B(c) := c \cap p \neq \emptyset$ where $p \subseteq W$, $o_{B,\emptyset}$ is distributive if and only if $p = \emptyset$ or $p = W$. (proof in the appendix)

[10]Note that by convention, $\bigcap \emptyset = W$, which is critical for the following conclusion.

[11]As it must be for every test-collapse that checks for compatibility, because compatibility tests, like $\diamond$ only return the input context when the intersection is non-empty.

Here's the idea. A substantive test is non-distributive because a union can pass a test that one of its parts fails, and the passing part will carry the failing part along. That rescue effect is available exactly when the test can discriminate between states. A test that delivers true for everything or false for everything can never produce a mismatch, and so is distributive. This is robustly important for van Benthem - distributivity in his definition is tracking whether a test discriminates between states. It delivers a verdict of static when the test has no content, and dynamic as soon as it has any.

5.2 Verdict Divergence

This contrast between vB and R&Y means that we can hold the system Simple Veltman developed fixed and vary only the proposition tested and the two sets of criteria come apart.

Consider a toy example where $W = \{w_1, w_2, w_3, w_4\}$, and an operator $o_{B,\emptyset}$, and three single-operation systems are defined differing only in which p is tested.

(23) p = $\emptyset$: B is uniformaly false, because o is the constant map to $\emptyset$. R&Y: static, vB: static.

(24) p = W: B is uniformly true where c is non-empty, and an empty c returns the empty state, so o is an identity mapping. R&Y: static, vB: static

(25) $p = w_1, w_2$: o is just SV's own $\diamond\phi$ for $[\![\phi]\!] = p$, which is a substantive test, and so is R&Y static, and vB dynamic.

The upshot here is that R&Y's diagnostic returns identical verdicts whether p is trivial or non-trivial. van Benthem returns opposite verdicts depending on substance.

6 This doesn't render Rothschild & Yalcin's criteria moot

I have shown that Y & R's characterization of dynamic systems departs from van Benthem's (1986). Using idempotence and commutivity doesn't merely allow generalization to non-intersective systems (while keeping the original van Benthem results for intersective systems intact) - it changes which *intersective* systems are characterized as dynamic.

vB's characterization builds upon key features of $\diamond$, in particular, to provide an upper bound on static conversation systems. In contrast, Simple Veltman shows that R&Y's characterization treats $\diamond$ as a dynamic operator insofar as it is *able to behave dynamically* in a conversation system, and provides at least a smaller upper bound on that set (if not a least upper bound) than vB does. This is because characterizations of dynamicness from Y&S result from the interaction of $\diamond$ with other updates. One way to demonstrate this, is

that a small enrichment of Simple Veltman renders the system non-commutative.

(26) **Facts+Checks**

L_{FC} Syntax

If ϕ is an atomic formula of L_{SV}, then ϕ is a WFF.

If ϕ is an atomic formula of L_{SV}, then $\ulcorner\diamond\phi\urcorner$ is a WFF.

L_{FC} Semantics

$$s[p] := s \cap [\![p]\!]$$

$$s[\diamond\phi] := \begin{cases} s \cap [\phi] \neq \emptyset & \text{return s} \\ s \cap [\phi] = \emptyset & \text{return } \emptyset \end{cases}$$

F+C increases the expressive power of SV only by way of increasing the number of possible output states from the initial context set. SV is non-commutative and hence dynamic for R&Y.

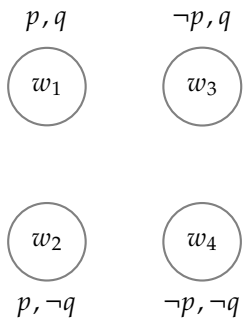


Figure 2: Possibilities for p and q

To make this concrete, consider $\{w_1, w_2, w_3, w_4\}$ again. Updating $\{w_1, w_3\}$[p] = $\{w_1\}$. This directly results in failures of commutativity. Take $\{w_2, w_3\}[q][\diamond p] = \emptyset$ as opposed to $\{w_2, w_3\}[\diamond p][q] = \{w_3\}$. This is because when updating with q and then $\diamond p$, q restricts the context set to $\{w_3\}$ which is incompatible with p and so returns the empty state. However, the initial state is compatible with p, and so updating with $\diamond p$ then q, returns the intial state and then intersects that set with the q-worlds, eliminating w_2 from the context set.

So, merely the addition of factual updates renders a testy-modal semantics dynamic both for vB and Y&S. This suggests that the two systems form a minimal pair. What R&Y's criteria are responding to is the addition of the clause and not anything about what $\diamond$-operator itself means. vB on the other hand was responding to what $\diamond$ meant all along. This can again be generalized.

(27) Let $o_1 = o_{B,K}$, and let o_2 be any operation with $Ko_2 = K$. Then o_1 and o_2 commute iff for every c, $B(c)$ and $B(co_2)$ agree, or $co_2 = K$.

Here's the idea. A test-collapse operator, so defined, is sensitive only to the verdict the test returns on it. So running another operation first can only make a difference if that other operation changes the verdict. And even a changed verdict is harmless if the other operation has deposited us at the fixed state, K. So the only genuinely disruptive move is one that flips the verdict while ending in a state other than k, which is a middle case where some content is removed, but not all of it.

Critically, this allows us to explicitly state that any two test-collapse operators commute unconditionally when they have the same failed fixed state.

(28) Any two operations $o_{B_1,K}$ and $o_{B_2,K}$ (same K) commute unconditionally, for an arbitrary B_1 and B_2.

This is because because test-collapse operation only the two moves that the previous proof allows: leave the state the same or send it to K. Since there is no third option, the tests can never disrupt each other, whatever they test. Facts+Checks demonstrates precisely what disrupts this. However, the change in judgment for R&Y on Facts+Checks is not dependent on factual update. Rather, what matters is how the two propositions being updated sit relative to each other. If one contains the other, then removing q-incompatible worlds can never eliminate the last of the p-worlds without emptying the state entirely, which is the harmless case we identified above. So, really what R&Y are sensitive to is whether each proposition has territory that the other lacks; whether one can build a state holding a p-world outside q as well as a q-world outisde of p. The formal result is this:

(29) Let $o_1 = O_{B,\emptyset}$ be the test on p and let $f_q(c) = c \cap q$ be an arbitrary intersective update. Then o_1 and f_q commute iff $p \subseteq q$ or $q \subseteq p$.

This result tells us that R&Y's criteria does discriminate. However, what it discriminates is whether the system's operations are of more than one kind, and when they are, how the propositions involved overlap. It never queries the content of an individual test.

6.1 Upshot

R&Y's diagnostic is blind along two independent axes, and van Benthem's is sensitive to both. PROOF X already makes this vivid. First, by X, idempotence holds for every K without exception, so R&Y's algebra

cannot distinguish an operation that collapses to the absurd state from one that, on failure, relocates to any other fixed state whatsoever; Y shows commutativity holds for any two test-collapse operations sharing a K, regardless of what that shared value is. van Benthem's criterion draws the distinction R&Y's cannot: by Z, eliminativity is a strong constraint on K, for any genuine compatibility test it forces $K = \emptyset$. An operation that collapses to some other, unmotivated state is unconditionally van-Benthem-dynamic, with no isomorphism available to rescue it. Second, distributivity (Z) further restricts which test predicates B may be used, while R&Y's diagnostic remains indifferent to the content of B. So within the entire test-collapse family, R&Y's criterion carries no information about either parameter B or K, while van Benthem's carries an abundance: it recognizes only the trivial-test corner of the space as static.

I have now shown that R&Y's criteria issues a judgment of staticness to the entire test-collapse family for single operator systems. The immediate question to ask is "What licenses such a cheap judgment?" In the following section, I argue that the answer is visible in R&Y's proof of the representation theorem.

7 Representation at a Higher Type

Recall that van Benthem's criteria is checked against the system's own set of worlds W. It requires that $p \subseteq W$ with $co = c \cup p$ for every c and doesn't allow relabeling. R&Y's strong staticness on the other hand is existentially quantified over any target system $\langle L, C\prime, \cdot[\![\cdot]\!]\prime \rangle$, via any bijection.

Their representation builds a witnessing system from a set h(c) defined by the states that c can be come. $cRc\prime$ (c reaches c$\prime$) just in case $c\prime$ is obtainable from c by finitely many applications of operations in O. In more formal terms:

(30) $h(c) := \{c\prime \in C : cRc\prime\}$ (the orbit of c); $j(o) := \{c \in C : co = c\}$ (the fixed points of o).

A low-brow way of describing what's going on is that h(c) is everything that a state c could still become, and j(o) is everything the operation o already leaves alone. Their central move then states that what c becomes under o is exactly the overlap of those two pieces: the future states available to c are also the static points for o. This is why their result deserves to be called intersective; they have literally rebuilt update from intersection.

Test collapse operators, with the same collapse state, guarantee this. once a context has passed a test, no further test can make it fail that test while leaving it alive because the only alternative a test offers is the absurd state.

Take an example. For Simple Veltman to count as static, $C = \wp(W)$: the witnessing worlds are type-1

objects (states) playing the role of type-0 primitives, and the witnessing system's states, being subsets of that world-set, are type 2.

Made fully explicit, for the single-operator instance already used above ($p = \{w_1, w_2\}, W = \{w_1, w_2, w_3, w_4\}$): a state c is compatible with p exactly when $c \cap p \neq \emptyset$, in which case $\diamond$ fixes c and idempotence closes the orbit, so $h(c) = c$; every incompatible c ($c \cap p = \emptyset$ has $\diamond$ send it to $\emptyset$, itself fixed, so $h(c) = c, \emptyset$. So the witnessing system's states are $C\prime = h[\wp(W)] = \{\{c\} : c \cap p \neq \emptyset \cup \{\{c, \emptyset\} : c \cap p = \emptyset\}$ — twelve states of the first shape (the twelve c compatible with p) and four of the second ($\emptyset, \{w_3\}, \{w_4\}, \{w_3, w_4\}$, the four incompatible with p). The single operation is intersection with $J := j(\diamond) = \{c : c \cap p \neq \emptyset\} \cup \emptyset$, the fixed points of $\diamond$. This is the concrete intersective system Simple Veltman's $\diamond$-operation is isomorphic to.

This is what the higher type is doing. A point of R&Y's witnessing system is a set of states. They are states of the original system which have been repackaged to serve as atoms of the new state. Specifically, since h: $C \to \wp(C)$, the witnessing system's worlds are elements of C itself.

7.1 Load-bearing or Cheap Representation Theorem?

van Benthem's conditions and Rothschild & Yalcin's ask the same question: can a system's updates be represented as intersection with a proposition? The main difference between them is how much freedom they allow the representing correspondence. Since literal intersection is itself idempotent and commutative, meeting either standard entails idempotence and commutativity. The difference between the two is whether than pair is enough to meet the standard, not whether its required. van Benthem fixes the correspondence to the identity on the system's own worlds (a highly strict setting). Whereas, R&Y allow any bijection onto any intersective system (a highly permissive setting). Neither is necessarily wrong, however R&Y occupy a permissive extreme without marking it as a choice, and the notion of staticness they are after doesn't force that choice. Simple Veltman sits where the two pull apart; it is dynamic by the strict standard, and static by the permissive one.

R&Y's construction, representing a structure by letting it act on its own parts, is a familiar move in mathematics. R&Y's operations generate a monoid - a set with an associative operation on identity[12]. Cayley's theorem (for group theory) states that every group embeds in a group of permutations of its own elements ($g \mapsto (x \mapsto gx)$). However, R&Y's construction is a monoid precisely because it lacks the invertibility of groups. Cayley's theorem for monoids states that every monoid M embeds, by $m \mapsto (x \mapsto mx)$, in the monoid of all functions from M to itself. This theorem is directly up R&Y's alley: build the representing points from the represented structure's own material, rather than reaching for anything

[12]Which is distinguished from a group in that it doesn't allow backpedalling. There is no guarantee that any operation can be undone.

independent.

Critical to my point is that the classical group theory construction already shows how cheap the move can be. S_3, the six permutations of three objects (e.g. {1,2,3}) has an obvious faithful action.[13] However, Cayley's construction ignores this and builds an action on six points (the group's own elements) permuted internally by left multiplication. Though the theorem is true, it is blind to the economical representation already available and defaults to inflating the domain to match the group's size.

Nonetheless, internal construction is not, by itself, what makes a representation theorem cheap. Stone's theorem utilizes ultrafilters which are told apart by nothing but an algebra's own elements. Distinct ultrafilters always disagree about some element of the algebra, so in this case an algebra's own elements separate every pair of points in the resulting space, making the initial algebra fully recoverable. The distinction between Stone's representation and Cayley's is whether the construction goes on to do further work: whether the points it produces are actually separated and the original structure is recoverable from the construction itself.

By that test, h[C] fails where Stone's ultrafilters succeed. The reason is a counting difference. A set of n distinguishing sets can tell apart amost 2^n classes of points because every point either belongs to a set or it doesn't. This means each point gets one of at most 2^n membership patterns across the n sets, and two points with the same pattern are indistinguishable by anything built out of those sets. R&Y's construction supplies |O| such distinguishing sets (from $\{j(o) : o \in O\}$. Since h is injective by R&Y's own result, h[C] has |C| points that need to be distinguished. Whenever C has more states than $2^{|O|}$ patterns to cover, some two of them must share a pattern.

Simple Veltman realizes this at its most extreme. One operation gives at most two patterns - inside $J = j(\diamond)$, or outside of it against sixteen states. Thirteen of those states (the twelve compatible states plust $\emptyset$, which is the fixed point of $\diamond$) shared the single pattern. and so are indistinguishable from one another by anything that either h or j provide.

This puts R&Y's representation theorem in Cayley's camp not Stone's because it is true, internally built, and provably too coarse to do anything other than certify the one bijection that it was built to produce.

8 Conclusion

[13]This is technically defined in group theory. An action of a group G on a set X is faithful when the only element in G that leaves every single element in X completely unchanged (fixed) is the identity element e.

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